

$$\textcircled{1} \quad 11110111_2$$

comp = complement;

ver = verification

$$= -128 + 64 + 32 + 16 + 4 + 2 + 1$$

$$= -128 + 119$$

$$= -9$$

ver:

$$-9$$

$$= 2's \text{ comp}(9)$$

$$= 2's \text{ comp}(00001001_2)$$

$$= 1's \text{ comp}(00001001_2) + 1$$

$$= 1110110_2 + 1$$

$$= 11110110_2$$

Python notes:

$$x = 0b1111011$$

print(x) # prints -9

$$y = -9$$

print(bin(y)) # prints '0b1111011'

$$\textcircled{2} 10111001_2$$

$$= -128 + 32 + 16 + 8 + 1$$

$$= -128 + 57$$

$$= -71$$

ver:

$$-71_{10}$$

$$= 2's \text{ comp}(71)$$

$$= 2's \text{ comp}(01000111_2)$$

$$= 1's \text{ comp}(01000111_2) + 1$$

$$= 10111000_2 + 1$$

$$= 10111001_2$$

$$\begin{array}{r} 1 \\ 32 \\ + 16 \\ 8 \\ 1 \\ \hline 57 \end{array}$$

$$\begin{array}{r} -1 \\ 128 \\ - 57 \\ \hline 071 \end{array}$$

$$\begin{array}{r} 2 \overline{) 71} \\ 2 \overline{) 35-1} \\ 2 \overline{) 17-1} \\ 2 \overline{) 8-1} \\ 2 \overline{) 4-0} \\ 2 \overline{) 2-0} \\ 2 \overline{) 1-0} \\ 2 \overline{) 1} \end{array}$$

$$\textcircled{3} \ 1110110_2$$

$$= -128 + 64 + 32 + 16 + 4 + 2$$

$$= -128 + 118$$

$$= -10$$

ver:

$$-10$$

$$= 2's \text{ comp}(10)$$

$$= 2's \text{ comp}(00001010_2)$$

$$= 1's \text{ comp}(00001010_2) + 1$$

$$= 1110101_2 + 1$$

$$= 1110110_2$$

$$\textcircled{4} \quad 1111101_2$$

$$= -128 + 64 + 32 + 16 + 8 + 4 + 1$$

$$= -128 + 125$$

$$= -3$$

ver:

-3

$$= 2's \text{ Comp}(3)$$

$$= 2's \text{ Comp}(00000011_2)$$

$$= 1's \text{ Comp}(00000011_2) + 1$$

$$= 1111100_2 + 1$$

$$= 1111101_2$$

$$\textcircled{5} \ 10010010_2$$

$$= -128 + 16 + 2$$

$$= -128 + 18$$

$$= -110$$

ver:

$$-110$$

$$= 2^1 \text{ comp } (110)$$

$$= 2^5 \text{ comp } (01101110_2)$$

$$= 1^5 \text{ comp } (01101110_2) + 1$$

$$= 10010001_2 + 1$$

$$= 10010010_2$$

$$\begin{array}{r} 2 \overline{) 110} \\ 2 \overline{) 550} \\ 2 \overline{) 27-1} \\ 2 \overline{) 13-1} \\ 2 \overline{) 6-1} \\ 2 \overline{) 3-0} \\ 2 \overline{) 1-1} \\ 1 \end{array}$$

$$\textcircled{6} 11010101_2$$

$$= -128 + 64 + 16 + 4 + 1$$

$$= -128 + 85$$

$$= -43$$

ver:

$$-43$$

$$= 2's \text{ comp}(43)$$

$$= 2's \text{ comp}(001010101_2)$$

$$= 1's \text{ comp}(001010101_2) + 1$$

$$= 11010100_2 + 1$$

$$= 11010101_2$$

$$\begin{array}{r} 64 \\ + 16 \\ + 4 \\ + 1 \\ \hline 85 \end{array}$$

$$\begin{array}{r} -128 \\ + 85 \\ \hline -43 \end{array}$$

$$\begin{array}{r} 2 \overline{) 43} \\ 2 \overline{) 21} -1 \\ 2 \overline{) 10} -1 \\ 2 \overline{) 5} -0 \\ 2 \overline{) 2} -1 \\ 2 \overline{) 1} -0 \\ 1 \end{array}$$

$$\textcircled{7} 1111111_2$$

$$= -128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

$$= -128 + 127$$

$$= -1$$

$$\begin{array}{r} 2 \\ 64 \\ 32 \\ + 16 \\ 8 \\ 4 \\ 2 \\ 1 \\ \hline 127 \end{array}$$

Ver:

-1

$$= 2's \text{ comp}(1)$$

$$= 2's \text{ comp}(00000001_2)$$

$$= 1's \text{ comp}(00000001_2) + 1$$

$$= 11111110_2 + 1$$

$$= 1111111_2$$

$$\textcircled{8} \ 1000000_2$$

$$= -128 + 0$$

$$= -128$$

ver:

$$-128$$

$$= 2's \text{ comp } (128)$$

$$= 2's \text{ comp } (1000000_2)$$

$$= 1's \text{ comp } (1000000_2) + 1$$

$$= 0111111_2 + 1$$

$$= 1000000_2$$

$$\textcircled{9} 11001001_2$$

$$= -128 + 64 + 8 + 1$$

$$= -128 + 73$$

$$= -55$$

ver!

$$-55$$

$$= 2's \text{ comp}(55)$$

$$= 2's \text{ comp}(0011011_2)$$

$$= 1's \text{ comp}(0011011_2) + 1$$

$$= 11001000_2 + 1$$

$$= 11001001$$

$$\begin{array}{r} \overset{-1}{1}28 \\ - \quad \quad 73 \\ \hline 055 \end{array} \quad \begin{array}{r} \overset{1}{6}4 \\ + \quad \quad 8 \\ \hline \quad \quad 73 \end{array}$$

$$\begin{array}{r} 2 \overline{) 55} \\ 2 \overline{) 27-1} \\ 2 \overline{) 13-1} \\ 2 \overline{) 6-1} \\ 2 \overline{) 3-0} \\ 2 \overline{) 1-1} \\ \hline 1 \end{array}$$

$$\textcircled{10} 11100111_2$$

$$= -128 + 64 + 32 + 4 + 2 + 1$$

$$= -128 + 103$$

$$= -25$$

Ver:

$$-25$$

$$= 2's \text{ comp}(25)$$

$$= 2's \text{ comp}(00011001_2)$$

$$= 1's \text{ comp}(00011001_2) + 1$$

$$= 11100110_2 + 1$$

$$= 11100111_2$$

$$\begin{array}{r} 64 \\ 32 \\ + 4 \\ 2 \\ 1 \\ \hline 103 \end{array}$$

$$\begin{array}{r} 128 \\ - 103 \\ \hline 025 \end{array}$$

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12} - 1 \\ 2 \overline{) 6} - 0 \\ 2 \overline{) 3} - 0 \\ 2 \overline{) 1} - 1 \\ \hline 1 \end{array}$$

$$\textcircled{11} -20$$

$$= 2's \text{ comp}(20)$$

$$= 2's \text{ comp}(00010100_2)$$

$$= 1's \text{ comp}(00010100_2) + 1$$

$$= 11101011_2 + 1$$

$$= 11101100_2$$

ver.

$$11101100_2$$

$$= -128 + 64 + 32 + 8 + 4$$

$$= -128 + 108$$

$$= -20$$

$$\begin{array}{r} 2 \overline{) 20} \\ 2 \overline{) 10} 0 \\ 2 \overline{) 5} 0 \\ 2 \overline{) 2} 1 \\ 2 \overline{) 1} 0 \\ 1 \end{array}$$

$$\begin{array}{r} 11101011 \\ + 1 \\ \hline 11101100 \end{array}$$

$$\begin{array}{r} 1 \\ 64 \\ 32 \\ + 8 \\ 4 \\ \hline 108 \end{array}$$

$$\begin{array}{r} 128 \\ - 108 \\ \hline 020 \end{array}$$

$$\textcircled{12} -96$$

$$= 2's \text{ comp}(96)$$

$$= 2's \text{ comp}(01100000_2)$$

$$= 1's \text{ comp}(01100000_2) + 1$$

$$= 10011111_2 + 1$$

$$= 10100000_2$$

ver:

$$10100000_2$$

$$= -128 + 32$$

$$= -96$$

$$\begin{array}{r} 2 \overline{) 96} \\ \underline{2 \overline{) 48} - 0} \\ 2 \overline{) 24} - 0 \\ \underline{2 \overline{) 12} - 0} \\ 2 \overline{) 6} - 0 \\ \underline{2 \overline{) 3} - 0} \\ 2 \overline{) 1} - 1 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 10011111 \\ + \\ \hline 10100000 \end{array}$$

$$\begin{array}{r} -1 \\ 128 \\ - 32 \\ \hline 096 \end{array}$$

$$\textcircled{13} -21$$

$$= 2's \text{ comp}(21)$$

$$= 2's \text{ comp}(00010101_2)$$

$$= 1's \text{ comp}(00010101_2) + 1$$

$$= 11101010_2 + 1$$

$$= 11101011_2$$

Ver:

$$11101011_2$$

$$= -128 + 64 + 32 + 8 + 2 + 1$$

$$= -128 + 107$$

$$= -21$$

$$\begin{array}{r} 2 \overline{) 21} \\ 2 \overline{) 10} - 1 \\ 2 \overline{) 5} - 0 \\ 2 \overline{) 2} - 1 \\ 2 \overline{) 1} - 0 \\ 1 \end{array}$$

$$\begin{array}{r} 1 \\ 64 \\ 32 \\ 8 \\ 2 \\ 1 \\ \hline 107 \end{array}$$

$$\begin{array}{r} 128 \\ - 107 \\ \hline 021 \end{array}$$

$$\textcircled{14} - 16$$

$$= 2^5 \text{ comp}(16)$$

$$= 2^5 \text{ comp}(00010000_2)$$

$$= 1^5 \text{ comp}(00010000_2) + 1$$

$$= 1110111_2 + 1$$

$$= 1110000_2$$

ver:

$$1110000_2$$

$$= -128 + 64 + 32 + 16$$

$$= -128 + 112$$

$$= -16$$

$$\begin{array}{r} 1110111 \\ + \\ \hline 1111000 \end{array}$$

$$\begin{array}{r} 1 \\ 64 \\ + 32 \\ + 16 \\ \hline 112 \end{array}$$

$$\begin{array}{r} 128 \\ - 112 \\ \hline 016 \end{array}$$

$$\textcircled{15} - 9$$

$$= 2's \text{ comp}(9)$$

$$= 2's \text{ comp}(00001001_2)$$

$$= 1's \text{ comp}(00001001_2) + 1$$

$$= 11110110_2 + 1$$

$$= 11110111_2$$

Ver:

$$11110111_2$$

$$= -128 + 64 + 32 + 16 + 4 + 2 + 1$$

$$= -128 + 119$$

$$= -9$$

$$\begin{array}{r} 64 \\ 32 \\ +16 \\ 4 \\ 2 \\ 1 \end{array}$$

$$\begin{array}{r} 128 \\ -119 \\ \hline 009 \end{array}$$

$$\textcircled{16} - 36$$

$$= 2's \text{ comp}(36)$$

$$= 2's \text{ comp}(00100100_2)$$

$$= 1's \text{ comp}(00100100_2) + 1$$

$$= 11011011_2 + 1$$

$$= 11011100$$

ver:

$$11011100_2$$

$$= -128 + 64 + 16 + 8 + 4$$

$$= -128 + 91$$

$$= -36$$

$$\begin{array}{r} 2 \overline{) 36} \\ \underline{2 18 - 0} \\ 2 9 - 0 \\ \underline{2 4 - 1} \\ 2 2 - 0 \\ \underline{2 1 - 0} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \\ + 64 \\ 16 \\ 8 \\ 4 \\ \hline 92 \end{array} \quad \begin{array}{r} -1 \\ 128 \\ - 92 \\ \hline 36 \end{array}$$

$$\textcircled{17} - 50$$

$$= 2's \text{ comp}(50)$$

$$= 2's \text{ comp}(00110010_2)$$

$$= 1's \text{ comp}(00110010_2) + 1$$

$$= 1100110_2 + 1$$

$$= 1100110_2$$

Ver:

$$1100110_2$$

$$= -128 + 64 + 8 + 4 + 2$$

$$= -128 + 78$$

$$= -50$$

$$2 \overline{) 50}$$

$$2 \overline{) 25-0}$$

$$2 \overline{) 12-1}$$

$$2 \overline{) 6-0}$$

$$2 \overline{) 3-0}$$

$$2 \overline{) 1-1}$$

1

1

64

+ 8

4

2

$$\underline{\hspace{1cm}} 78$$

-128

- 78

50

$$\textcircled{18} - 101$$

$$= 2's \text{ comp}(101)$$

$$= 2's \text{ comp}(01100101_2)$$

$$= 1's \text{ comp}(01100101_2) + 1$$

$$= 10011010_2 + 1$$

$$= 10011011_2$$

ver:

$$10011011_2$$

$$= -128 + 16 + 8 + 2 + 1$$

$$= -128 + 27$$

$$= -101$$

$$\begin{array}{r} 2 \overline{) 101} \\ 2 \overline{) 50-1} \\ 2 \overline{) 25-0} \\ 2 \overline{) 12-1} \\ 2 \overline{) 6-0} \\ 2 \overline{) 3-0} \\ 2 \overline{) 1-1} \\ 1 \end{array}$$

$$\begin{array}{r} 1 \\ + 16 \\ + 8 \\ + 2 \\ + 1 \\ \hline 27 \end{array} \quad \begin{array}{r} 128 \\ - 27 \\ \hline 101 \end{array}$$

$$\textcircled{19} - 123$$

$$= 2's \text{ comp}(123)$$

$$= 2's \text{ comp}(01111011_2)$$

$$= 1's \text{ comp}(01111011_2) + 1$$

$$= 10000100_2 + 1$$

$$= 10000101_2$$

ver:

$$10000101_2$$

$$= -128 + 4 + 1$$

$$= -128 + 5$$

$$= -123$$

$$\begin{array}{r} 2 \overline{) 123} \\ \underline{2 \overline{) 61-1}} \\ 2 \overline{) 30-1} \\ \underline{2 \overline{) 15-0}} \\ 2 \overline{) 7-1} \\ \underline{2 \overline{) 3-1}} \\ 2 \overline{) 1-1} \\ \hline 1 \end{array}$$

$$\textcircled{20} - 5$$

$$= 2's \text{ Comp}(5)$$

$$= 2's \text{ Comp}(00000101_2)$$

$$= 1's \text{ Comp}(00000101_2) + 1$$

$$= 1111010 + 1$$

$$= 1111011_2$$

ver:

$$1111011_2$$

$$= -128 + 64 + 32 + 16 + 8 + 2 + 1$$

$$= -128 + 123$$

$$= -5$$

	2	
	64	- 128
	32	123
+	16	<u>005</u>
	8	
	2	
	1	
	<u>123</u>	